HDOC- 9 – 1
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Program to Assign a Grade Based on Percentage Using if–else

This document provides a clean, well-commented implementation for assigning a grade from a percentage in C. The program accepts a percentage between 0 and 100, validates it, and uses an if–else if–else ladder to assign Grade A, B, C, D or F.

1. C Implementation

Step 1: Understand the Logic

- 90 to 100: Grade A | 80 to 89: Grade B | 70 to 79: Grade C | 60 to 69: Grade D | below 60: Grade F.
- The conditions are checked from the highest grade downwards, so each condition only needs a lower limit (for example percentage >= 80), because values of 90 or more were already handled by the earlier condition.
- A float variable is used so that percentages with decimals (for example 89.5) are graded correctly.
- A value below 0 or above 100 is not a valid percentage, so it is rejected before grading.

Step 2: Define Variables and Data Types

- Use a float variable to store the percentage.

Step 3: Take User Input

- Display a prompt using printf("Enter percentage (0-100): "), then read the value using scanf("%f", &percentage).

Step 4: Validate the Input

- If percentage < 0 or percentage > 100, print an error message and do not assign a grade.

Step 5: Assign the Grade Using the if–else if Ladder

1. If percentage >= 90, print Grade A.
2. Else if percentage >= 80, print Grade B.
3. Else if percentage >= 70, print Grade C.
4. Else if percentage >= 60, print Grade D.
5. Otherwise (else), print Grade F.

Complete C Code

```c
/*
 * Q13 (User Inputs, Operations & Output)
 * Write a program that accepts a percentage (0-100) and assigns
 * a grade: 90-100 A, 80-89 B, 70-79 C, 60-69 D, below 60 F.
 */
#include <stdio.h>

int main() {
    float percentage;

    // Step 1: Read the percentage
    printf("Enter percentage (0-100): ");
    scanf("%f", &percentage);

    // Step 2: Validate the range
    if (percentage < 0 || percentage > 100) {
        printf("Invalid percentage! Enter a value between 0 and 100.\n");
    } else {
        // Step 3: Assign the grade
        if (percentage >= 90) {
            printf("Grade: A\n");
        } else if (percentage >= 80) {
            printf("Grade: B\n");
        } else if (percentage >= 70) {
            printf("Grade: C\n");
        } else if (percentage >= 60) {
            printf("Grade: D\n");
        } else {
            printf("Grade: F\n");
        }
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc9_1.c -o hdoc9_1
$ ./hdoc9_1
Enter percentage (0-100): 95
Grade: A
```

Run 2:

```
$ ./hdoc9_1
Enter percentage (0-100): 85
Grade: B
```

Run 3:

```
$ ./hdoc9_1
Enter percentage (0-100): 72
Grade: C
```

Run 4:

```
$ ./hdoc9_1
Enter percentage (0-100): 65
Grade: D
```

Run 5:

```
$ ./hdoc9_1
Enter percentage (0-100): 45
Grade: F
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 95
- Calculation:
  - Validation: 95 < 0 or 95 > 100 is FALSE, so the input is valid
  - 95 >= 90 is TRUE, so Grade A is assigned
- Output: Grade: A

Sample Test Case 2:

- Input: 85
- Calculation:
  - 85 >= 90 is FALSE
  - 85 >= 80 is TRUE, so Grade B is assigned
- Output: Grade: B

Sample Test Case 3:

- Input: 72
- Calculation:
  - 72 >= 90 is FALSE and 72 >= 80 is FALSE
  - 72 >= 70 is TRUE, so Grade C is assigned
- Output: Grade: C

Sample Test Case 4:

- Input: 65
- Calculation:
  - 65 >= 90, 65 >= 80 and 65 >= 70 are all FALSE
  - 65 >= 60 is TRUE, so Grade D is assigned
- Output: Grade: D

Sample Test Case 5:

- Input: 45
- Calculation:
  - 45 >= 90, 45 >= 80, 45 >= 70 and 45 >= 60 are all FALSE
  - The else block runs, so Grade F is assigned
- Output: Grade: F